C6.2

Continuous Optimization

Lecture notes

(based on the lectures of Coralia Cartis, Oxford)

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Chapter 1

Optimality Conditions for Unconstrained Optimization

1.1 Problem classes and notation

We consider the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x), \tag{UP}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is smooth (typically $f \in \mathcal{C}^1$ or $\mathcal{C}^2$).

Definition 1.1 (Local and global minimizers)**.** A point $x^* \in \mathbb{R}^n$ is a *local minimizer* of f if there exists $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in N(x^*, \delta) = \{x : \|x - x^*\| < \delta\}$. It is a *strict* local minimizer if the inequality is strict for $x \neq x^*$, and a *global minimizer* if $f(x^*) \leq f(x)$ for all $x \in \mathbb{R}^n$.

Definition 1.2 (Descent direction)**.** Given $x \in \mathbb{R}^n$ with $\nabla f(x) \neq 0$, a vector $s \in \mathbb{R}^n$ is a *descent direction* for f at x if $\nabla f(x)^\top s < 0$.

Lemma 1.3 (Descent property)**.** *If $f \in \mathcal{C}^1$ and s is a descent direction at x, then there exists $\bar{\alpha} > 0$ such that $f(x + \alpha s) < f(x)$ for all $\alpha \in (0, \bar{\alpha}]$.*

Proof. Since s is a descent direction, $\nabla f(x)^\top s < 0$. By continuity of ∇f, there exists $\bar{\alpha} > 0$ such that

$$\nabla f(x + ts)^\top s < 0$$

for all $t \in [0, \bar{\alpha}]$. Hence, for any $\alpha \in (0, \bar{\alpha}]$, by the mean value theorem there exists $t \in (0, \alpha)$ such that

$$f(x + \alpha s) - f(x) = \alpha \cdot \nabla f(x + ts)^\top s < 0.$$

□

1.2 First-order optimality conditions

Theorem 1.4 (First-order necessary conditions)**.** *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$. If x^* is a local minimizer of f, then $\nabla f(x^*) = 0$.*

Proof. Suppose not. Put $s = -\nabla f(x^*)$ and note that s is a descent direction. By Lemma 1.3 there exists $\bar{\alpha} > 0$ such that $f(x^* + \alpha s) < f(x^*)$ for all $\alpha \in (0, \bar{\alpha}]$, contradiction. □

Definition 1.5 (Stationary point)**.** A point x with $\nabla f(x) = 0$ is called a *stationary point* of f.

1.3 Second-order optimality conditions

Theorem 1.6 (Second-order necessary conditions)**.** *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$. If x^* is a local minimizer of f, then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succeq 0$ (positive semidefinite), i.e. $s^\top \nabla^2 f(x^*) s \geq 0$ for all $s \in \mathbb{R}^n$.*

Proof. By Theorem 1.4, $\nabla f(x^*) = 0$. Suppose for contradiction that $s^\top \nabla^2 f(x^*) s < 0$ for some $s \in \mathbb{R}^n$. Since $s \neq 0$ and $f \in \mathcal{C}^2$, there exists $\hat{\alpha} > 0$ such that $s^\top \nabla^2 f(x^* + \alpha s) s < 0$ for all $\alpha \in (0, \hat{\alpha}]$. By Taylor's theorem, for any $\alpha \in (0, \hat{\alpha}]$, there exists $\bar{\alpha} \in (0, \alpha)$ such that

$$f(x^* + \alpha s) - f(x^*) = \frac{\alpha^2}{2} s^\top \nabla^2 f(x^* + \bar{\alpha} s) s < 0,$$

contradiction. □

Theorem 1.7 (Second-order sufficient conditions)**.** *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$. If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$ (positive definite), then x^* is a strict local minimizer of f.*

Proof. Since $f \in \mathcal{C}^2$, there exists $\hat{\alpha} > 0$ such that $s^\top \nabla^2 f(x^* + \alpha s) s > 0$ for all $\alpha \in (0, \hat{\alpha}]$ and $s \in \mathbb{R}^n$. By Taylor's theorem, for any $\alpha \in (0, \hat{\alpha}]$, there exists $\bar{\alpha} \in (0, \alpha)$ such that

$$f(x^* + \alpha s) - f(x^*) = \frac{\alpha^2}{2} s^\top \nabla^2 f(x^* + \bar{\alpha} s) s > 0.$$

Thus x^* is a strict local minimizer within the ball $N(x^*, \hat{\alpha})$. □

1.4 Convex problems

Definition 1.8 (Convex function)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is *convex* if its domain is convex and, for all x, y in its domain and all $\lambda \in [0, 1]$, $f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y)$.

Theorem 1.9 (Optimality for convex problems)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and $\mathcal{C}^1$. Then x^* is a global minimizer of f if and only if $\nabla f(x^*) = 0$.*

Proof. If x^* is a global minimizer, then by Theorem 1.4, $\nabla f(x^*) = 0$. It remains to show the converse.

We first show that if x^* is a local minimizer, then it is a global minimizer. Suppose not. There exists $y \in \mathbb{R}^n$ such that $f(y) < f(x^*)$. Put $s = y - x^*$. By convexity, for $\lambda \in (0, 1)$,

$$f(x^* + \lambda s) = f((1 - \lambda) x^* + \lambda y) \leq (1 - \lambda) f(x^*) + \lambda f(y) < f(x^*).$$

But then x^* is a local minimizer, so there exists $\delta > 0$ such that $f(x^* + \lambda s) \leq f(x^*)$ for all $\lambda \in (0, \delta)$, contradiction.

[Proof to be filled in.] □

Chapter 2

Linesearch Methods

2.1 The generic linesearch method

Algorithm 2.1 (Generic Linesearch Method (GLM))**.** Choose $\varepsilon > 0$ and $x^0 \in \mathbb{R}^n$. For $k \geq 0$:

1. If $\left\|\nabla f(x^k)\right\| \leq \varepsilon$, stop.
2. Compute a descent direction s^k with $\nabla f(x^k)^\top s^k < 0$.
3. Compute a stepsize $\alpha_k > 0$ along s^k such that $f(x^k + \alpha_k s^k) < f(x^k)$.
4. Set $x^{k+1} := x^k + \alpha_k s^k$.

2.2 Stepsize rules: the Armijo condition

For $\beta \in (0,1)$ the *Armijo (sufficient decrease) condition* at iteration k is

$$f(x^k + \alpha s^k) \leq f(x^k) + \beta\alpha\, \nabla f(x^k)^\top s^k. \tag{ac}$$

Algorithm 2.2 (Backtracking-Armijo (bArmijo) linesearch)**.** Choose $\alpha^{(0)} > 0$, $\tau \in (0,1)$, $\beta \in (0,1)$. Set $i := 0$. While (ac) fails with $\alpha = \alpha^{(i)}$, set $\alpha^{(i+1)} := \tau\alpha^{(i)}$ and $i := i+1$. Set $\alpha_k := \alpha^{(i)}$.

Lemma 2.3 (Range of admissible stepsizes)**.** *Let $f \in \mathcal{C}^1$ with ∇f Lipschitz continuous with constant L. At iteration k of GLM, the Armijo condition* (ac) *is satisfied for all $\alpha \in [0, \alpha^k_{\max}]$, where*

$$\alpha^k_{\max} = \frac{(\beta - 1)\, \nabla f(x^k)^\top s^k}{L \left\|s^k\right\|^2}.$$

Proof. First note that $\alpha^k_{\max} < 0$ as $\beta < 1$ and s^k is a descent direction. Let s^k be a descent direction and let $\alpha \in [0, \alpha^k_{\max}]$. By Taylor's theorem, there exists $\tilde{\alpha} \in (0, \alpha)$ such that

$$\begin{aligned}
f(x^k + \alpha s^k) &= f(x^k) + \alpha \nabla f(x^k + \tilde{\alpha} s^k)^\top s^k \\
&= f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha \left[\nabla f(x^k + \tilde{\alpha} s^k) - \nabla f(x^k)\right]^\top s^k \\
&\leq f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha L \left\|x^k - \tilde{\alpha} s^k - x^k\right\| \cdot \left\|s^k\right\| \\
&\leq f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha^2 L \left\|s^k\right\|^2 \\
&\leq f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha^2 (\beta - 1) \nabla f(x^k)^\top s^k \\
&\leq f(x^k) + \beta\alpha \nabla f(x^k)^\top s^k,
\end{aligned}$$

which satisfies the Armijo condition. □

Lemma 2.4 (Lower bound on bArmijo stepsize)**.** *Under the conditions of Lemma 2.3, the bArmijo stepsize α_k satisfies $\alpha_k \geq \min\{\alpha^{(0)}, \tau\alpha^k_{\max}\}$ for all $k \geq 0$.*

Proof. Note that $a_k = a^{(0)}$ if bArmijo terminates at the start. Suppose this is not the case. By Lemma 2.3, bArmijo terminates whenever $\alpha \leq \alpha^k_{\max}$. Let $i-1$ be the last iteration before termination, so $\alpha^{(i-1)} > \alpha^k_{\max}$ and $\tau\alpha^{(i-1)} = \alpha^{(i)} \leq \alpha^k_{\max}$. But then

$$\alpha_k = \alpha^{(i)} = \tau\alpha^{(i-1)} > \tau\alpha^k_{\max}.$$

□

2.3 Global convergence of GLM

Theorem 2.5 (Global convergence of GLM with bArmijo)**.** *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$ be bounded below by f_{low}, and ∇f be Lipschitz continuous. Apply GLM with bArmijo linesearch ($\varepsilon := 0$). Then either there exists $\ell \geq 0$ with $\nabla f(x^\ell) = 0$, or*

$$\lim_{k\to\infty} \min\left\{\frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|}, |\nabla f(x^k)^\top s^k|\right\} = 0.$$

Proof. Suppose $\nabla f(x^k) \neq 0$ for all $k \geq 0$. Then Armijo condition with $x^{k+1} = x^k + \alpha_k s^k$ gives

$$f(x^{k+1}) \leq f(x^k) + \beta\alpha_k \nabla f(x^k)^\top s^k \implies f(x^k) - f(x^{k+1}) \geq \beta\alpha_k |\nabla f(x^k)^\top s^k|.$$

for all $k \geq 0$. By the telescoping sum,

$$f(x^0) - f(x^{k+1}) \geq \sum_{i=0}^{k} \beta\alpha_i |\nabla f(x^i)^\top s^i|.$$

Since f is bounded below by f_{low},

$$\infty > f(x^0) - f_{\text{low}} \geq \sum_{i=0}^{\infty} \beta\alpha_i |\nabla f(x^i)^\top s^i|.$$

Thus, $\alpha_k|\nabla f(x^k)^\top s^k| \to 0$ as $k \to \infty$. Recall from Lemma 2.4 that $\alpha_k \geq \min\{\alpha^{(0)}, \tau\alpha^k_{\max}\}$ for all $k \geq 0$. Define

$$\mathcal{K}_1 := \{k : \alpha^{(0)} \geq \tau\alpha^k_{\max}\}, \qquad \mathcal{K}_2 := \{k : \alpha^{(0)} < \tau\alpha^k_{\max}\}.$$

By Lemma 2.4, for $k \in \mathcal{K}_1$,

$$\alpha_k |\nabla f(x^k)^\top s^k| \geq \frac{(1-\beta)\tau}{L} \cdot \left(\frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|}\right)^2 \geq 0.$$

But then $\alpha_k|\nabla f(x^k)^\top s^k| \to 0$ as $k \to \infty$ in $\mathcal{K}_1$, so

$$\lim_{\substack{k\to\infty \\ k\in\mathcal{K}_1}} \frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|} \to 0.$$

On the other hand, by Lemma 2.4, for $k \in \mathcal{K}_2$ we have $a_k \geq \alpha^{(0)}$, so

$$0 \leq \lim_{\substack{k\to\infty \\ k\in\mathcal{K}_2}} |\nabla f(x^k)^\top s^k| \leq \frac{1}{\alpha^{(0)}} \cdot \lim_{\substack{k\to\infty \\ k\in\mathcal{K}_2}} \alpha_k |\nabla f(x^k)^\top s^k| \to 0.$$

Combining both cases, we have

$$\lim_{k\to\infty} \min\left\{\frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|}, \nabla f(x^k)^\top s^k|\right\} = 0.$$

□

Remark 2.6. Writing $\cos\theta_k = \frac{-\nabla f(x^k)^\top s^k}{\|\nabla f(x^k)\|\|s^k\|}$, Theorem 2.5 gives $\left\|\nabla f(x^k)\right\| \cos\theta_k \min\{1, \left\|s^k\right\|\} \to 0$. Hence to ensure $\left\|\nabla f(x^k)\right\| \to 0$ it is not enough that s^k be descent: the angle condition $\cos\theta_k \geq \delta > 0$ for some δ is required.

Chapter 3

The Steepest Descent Method

3.1 The steepest descent direction

Definition 3.1 (Steepest descent direction)**.** The *steepest descent (SD) direction* at x^k is $s^k_{\mathrm{SD}} := -\nabla f(x^k)$.

Lemma 3.2 (Optimality of $-\nabla f$)**.** *The steepest descent direction is the unique global solution of*

$$\min_{s\in\mathbb{R}^n} \; f(x^k) + s^\top \nabla f(x^k) \quad \textit{subject to} \quad \|s\| = \left\|\nabla f(x^k)\right\|.$$

Proof. Note that

$$\arg\min_{s\in\mathbb{R}^n} f(x^k) + s^\top \nabla f(x^k) = \arg\min_{s\in\mathbb{R}^n} s^\top \nabla f(x^k).$$

By Cauchy-Schwarz inequality,

$$|s^\top \nabla f(x^k)| \le \|s\| \left\|\nabla f(x^k)\right\| = \left\|\nabla f(x^k)\right\|^2,$$

with equality if and only if $s = \pm\nabla f(x^k)$. In particular,

$$s^\top \nabla f(x^k) \ge -\left\|\nabla f(x^k)\right\|^2,$$

which is only achieved when $s = -\nabla f(x^k)$. □

3.2 Global convergence of steepest descent

Theorem 3.3 (Global convergence of SD with bArmijo)**.** *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$ be bounded below with ∇f Lipschitz continuous. Apply GLM with $s^k = -\nabla f(x^k)$ and bArmijo linesearch. Then either there exists ℓ with $\nabla f(x^\ell) = 0$, or $\left\|\nabla f(x^k)\right\| \to 0$ as $k \to \infty$.*

Proof. Suppose there are no ℓ with $\nabla f(x^\ell) = 0$. Then by Theorem 2.5,

$$\lim_{k\to\infty} \min\left\{ \frac{\left\|\nabla f(x^k)\right\|^2}{\|-\nabla f(x^k)\|}, \left\|\nabla f(x^k)\right\| \right\} = \left\|\nabla f(x^k)\right\| = 0.$$

□

3.3 Local rate of convergence

Definition 3.4 (Rates of convergence)**.** A sequence $\{x^k\} \subset \mathbb{R}^n$ converges to x^* with rate $p \geq 1$ if there exist $\rho > 0$ and k_0 with $\left\|x^{k+1} - x^*\right\| \leq \rho \left\|x^k - x^*\right\|^p$ for all $k \geq k_0$. The cases $p = 1$ ($\rho < 1$) and $p = 2$ are called *linear* and *quadratic* convergence respectively. The convergence is *superlinear* if $\left\|x^{k+1} - x^*\right\| / \left\|x^k - x^*\right\| \to 0$.

Theorem 3.5 (Local linear convergence of SD)**.** *Let $f \in \mathcal{C}^2$ and let x^* be a local minimizer of f with $\nabla^2 f(x^*) \succ 0$ and eigenvalues $0 < \lambda^*_{\min} \leq \lambda^*_{\max}$. Apply SD with exact linesearch. If $x^k \to x^*$, then the convergence is linear with factor*

$$\rho \leq \frac{\kappa(x^*) - 1}{\kappa(x^*) + 1}, \qquad \kappa(x^*) = \frac{\lambda^*_{\max}}{\lambda^*_{\min}}.$$

Proof. [Proof to be filled in.] □

Chapter 4

Newton's Method

4.1 The Newton direction

Definition 4.1 (Newton direction)**.** For $f \in \mathcal{C}^2$ with $\nabla^2 f(x^k)$ nonsingular, the *Newton direction* at x^k is the solution s^k of

$$\nabla^2 f(x^k)\, s^k = -\nabla f(x^k). \tag{4.1}$$

The *(pure) Newton iteration* sets $x^{k+1} := x^k + s^k$.

Lemma 4.2 (Descent property of the Newton direction)**.** *If $\nabla^2 f(x^k) \succ 0$, then the Newton direction is a descent direction. Moreover, s^k is the unique minimizer of the second-order Taylor model $q_k(s) := f(x^k) + s^\top \nabla f(x^k) + \frac{1}{2} s^\top \nabla^2 f(x^k) s$.*

Proof. [Proof to be filled in.] □

4.2 Local convergence of pure Newton

Theorem 4.3 (Local quadratic convergence of pure Newton)**.** *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$, $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ nonsingular, and $\nabla^2 f$ locally Lipschitz continuous at x^*. If x^{k_0} is sufficiently close to x^* for some $k_0 \geq 0$, then x^k is well-defined for all $k \geq k_0$ and $x^k \to x^*$ at a quadratic rate.*

Proof. [Proof to be filled in.] □

4.3 Linesearch Newton (damped Newton) method

Algorithm 4.4 (Linesearch Newton method)**.** Choose $\varepsilon > 0$ and $x^0 \in \mathbb{R}^n$. While $\left\|\nabla f(x^k)\right\| > \varepsilon$: solve $\nabla^2 f(x^k) s^k = -\nabla f(x^k)$, then set $x^{k+1} = x^k + \alpha_k s^k$ with $\alpha_k \in (0, 1]$ chosen by, e.g., bArmijo linesearch.

Theorem 4.5 (Local convergence of linesearch Newton)**.** *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$ with $\nabla^2 f$ Lipschitz continuous and positive definite at the iterates. Apply Newton's method with bArmijo linesearch using $\beta \leq 1/2$ and $\alpha^{(0)} = 1$. If $x^k \to x^*$ with $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$, then $\alpha_k = 1$ for all k sufficiently large and the convergence is asymptotically quadratic.*

Proof. [Proof to be filled in.] □

4.4 Global convergence with general second-order models

In GLM, let B_k be symmetric positive definite for all k, and let s^k solve

$$B_k s^k = -\nabla f(x^k). \tag{4.2}$$

Theorem 4.6 (Global convergence with bounded B_k)**.** *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$ be bounded below with ∇f Lipschitz continuous. Apply GLM with s^k from* (4.2) *and bArmijo linesearch ($\varepsilon := 0$). Suppose the eigenvalues of B_k are uniformly bounded above and uniformly bounded away from zero, independently of k. Then either there exists ℓ with $\nabla f(x^\ell) = 0$, or $\left\|\nabla f(x^k)\right\| \to 0$ as $k \to \infty$.*

Proof. [Proof to be filled in.] $\square$

Chapter 5

Quasi-Newton Methods and Nonlinear Least-Squares

5.1 The secant equation

In quasi-Newton methods, $\nabla^2 f(x^{k+1})$ is approximated by a symmetric matrix B_{k+1} satisfying the *secant equation*

$$B_{k+1}\,\delta^k = \gamma^k, \qquad \delta^k := x^{k+1} - x^k, \;\; \gamma^k := \nabla f(x^{k+1}) - \nabla f(x^k). \tag{5.1}$$

Lemma 5.1 (Quadratic case)**.** *If $f(x) = g^\top x + \frac{1}{2}x^\top Hx$, then* (5.1) *is satisfied with $B_{k+1} = \nabla^2 f = H$.*

Proof. [Proof to be filled in.] □

5.2 The BFGS update

Definition 5.2 (BFGS update)**.** The *BFGS* update is the rank-2 symmetric update

$$B_{k+1} = B_k - \frac{B_k\delta^k(\delta^k)^\top B_k}{(\delta^k)^\top B_k\delta^k} + \frac{\gamma^k(\gamma^k)^\top}{(\delta^k)^\top\gamma^k}. \tag{5.2}$$

Theorem 5.3 (Properties of the BFGS update)**.** *If B_k is symmetric positive definite and $(\delta^k)^\top\gamma^k > 0$, then the BFGS update* (5.2) *is well-defined, symmetric and positive definite, and satisfies the secant equation* (5.1)*.*

Proof. [Proof to be filled in.] □

Remark 5.4. The condition $(\delta^k)^\top\gamma^k > 0$ is guaranteed by use of the Wolfe linesearch.

Theorem 5.5 (Convergence of BFGS)**.** *Under suitable conditions on f and using the Wolfe linesearch, the BFGS method is globally convergent and locally Q-superlinearly convergent.*

Proof. [Proof to be filled in.] □

5.3 Nonlinear least-squares

The nonlinear least-squares problem is

$$\min_{x\in\mathbb{R}^n} \; f(x) = \tfrac{1}{2}\,\|r(x)\|^2 = \tfrac{1}{2}\sum_{j=1}^{m} r_j(x)^2, \tag{5.3}$$

where $r : \mathbb{R}^n \to \mathbb{R}^m$ is smooth with Jacobian $J(x)$.

Lemma 5.6 (Derivatives of f)**.** *For f in* (5.3)*,*

$$\nabla f(x) = J(x)^\top r(x), \qquad \nabla^2 f(x) = J(x)^\top J(x) + \sum_{j=1}^{m} r_j(x) \nabla^2 r_j(x).$$

Proof. [Proof to be filled in.] □

5.4 The Gauss-Newton method

Definition 5.7 (Gauss-Newton direction)**.** The *Gauss-Newton (GN) direction* solves the linear system

$$J(x^k)^\top J(x^k)\, s^k = -J(x^k)^\top r(x^k). \tag{5.4}$$

Lemma 5.8 (Descent property)**.** *If $J(x^k)$ has full column rank, then the Gauss-Newton direction is a descent direction for f.*

Proof. [Proof to be filled in.] □

Theorem 5.9 (Convergence of Gauss-Newton)**.** *Apply Gauss-Newton with bArmijo linesearch.*

1. *(Global.) If $J(x^k)$ is uniformly full rank, $f \in \mathcal{C}^1$ bounded below and ∇f Lipschitz, then $\left\|\nabla f(x^k)\right\| \to 0$.*
2. *(Local.) If $r(x^*) = 0$ and $J(x^*)$ has full rank, the convergence is quadratic; if $r(x^*) \neq 0$ but small, the convergence is linear.*

Proof. [Proof to be filled in.] □

Chapter 6

Trust-Region Methods

6.1 The trust-region framework

At iteration k, we form a quadratic model

$$m_k(s) = f(x^k) + s^\top \nabla f(x^k) + \tfrac{1}{2} s^\top B_k s, \tag{6.1}$$

where B_k is symmetric (e.g. $B_k = \nabla^2 f(x^k)$). The *trust-region (TR) subproblem* is

$$\min_{s \in \mathbb{R}^n} \; m_k(s) \quad \text{subject to} \quad \|s\| \le \Delta_k, \tag{TR}$$

with trust-region radius $\Delta_k > 0$.

Algorithm 6.1 (Generic Trust-Region (GTR) method)**.** Given $\Delta_0 > 0$, $x^0 \in \mathbb{R}^n$, $\varepsilon > 0$. While $\left\|\nabla f(x^k)\right\| > \varepsilon$:

1. Form m_k and (approximately) solve (TR) for s^k with $m_k(s^k) < f(x^k)$.
2. Compute $\rho_k = \dfrac{f(x^k) - f(x^k + s^k)}{f(x^k) - m_k(s^k)}$.
3. If $\rho_k \ge 0.9$: set $x^{k+1} := x^k + s^k$, $\Delta_{k+1} := 2\Delta_k$.
   Else if $\rho_k \ge 0.1$: set $x^{k+1} := x^k + s^k$, $\Delta_{k+1} := \Delta_k$.
   Else: set $x^{k+1} := x^k$, $\Delta_{k+1} := \frac{1}{2}\Delta_k$.

6.2 The Cauchy point

Definition 6.2 (Cauchy point)**.** The *Cauchy point* of (TR) is $y_c^k = x^k + s_c^k$, where $s_c^k = -\alpha_c^k \nabla f(x^k)$ and

$$\alpha_c^k = \operatorname*{arg\,min}_{\alpha > 0} \; m_k\Big(-\alpha \nabla f(x^k)\Big) \quad \text{subject to} \quad \left\|\alpha \nabla f(x^k)\right\| \le \Delta_k.$$

Lemma 6.3 (Cauchy-point decrease)**.** *With* $h_k = \nabla f(x^k)^\top B_k \nabla f(x^k)$,

$$f(x^k) - m_k(s_c^k) \ge \tfrac{1}{2} \left\|\nabla f(x^k)\right\| \min\left\{\Delta_k, \frac{\left\|\nabla f(x^k)\right\|}{\|B_k\|}\right\}.$$

Proof. [Proof to be filled in.] □

6.3 Global convergence of GTR

Theorem 6.4 (Global convergence of GTR)**.** *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$ be bounded below with ∇f Lipschitz continuous, and assume $\|B_k\|$ is uniformly bounded. Apply GTR (Algorithm 6.1) requiring $m_k(s^k) \leq m_k(s_c^k)$. Then either there exists $\ell \geq 0$ with $\nabla f(x^\ell) = 0$, or $\liminf_{k\to\infty} \left\|\nabla f(x^k)\right\| = 0$.*

Proof. [Proof to be filled in.] □

Theorem 6.5 (Stronger global convergence)**.** *Under the conditions of Theorem 6.4, $\lim_{k\to\infty} \left\|\nabla f(x^k)\right\| = 0$.*

Proof. [Proof to be filled in.] □

6.4 Solving the trust-region subproblem

Theorem 6.6 (Characterization of the global TR solution)**.** *A vector $s^* \in \mathbb{R}^n$ is a global solution of* (TR) *if and only if $\|s^*\| \leq \Delta_k$ and there exists $\lambda^* \geq 0$ such that*

1. $(B_k + \lambda^* I)s^* = -\nabla f(x^k)$,
2. $\lambda^*(\Delta_k - \|s^*\|) = 0$ *(complementarity),*
3. $B_k + \lambda^* I \succeq 0$.

Moreover λ^* is unique whenever $B_k + \lambda^* I \succ 0$.

Proof. [Proof to be filled in.] □

Chapter 7

Optimality Conditions for Constrained Optimization

7.1 The constrained problem

We consider the general constrained problem

$$\min_{x\in\mathbb{R}^n} f(x) \quad \text{subject to} \quad c_i(x) = 0,\ i \in E;\ \ c_i(x) \geq 0,\ i \in I, \tag{CP}$$

with feasible set $\Omega = \{x \in \mathbb{R}^n : c_i(x) = 0,\ i \in E;\ c_i(x) \geq 0,\ i \in I\}$, and active set at $x \in \Omega$: $\mathcal{A}(x) = E \cup \{i \in I : c_i(x) = 0\}$.

7.2 Tangent cone and linearized feasible directions

Definition 7.1 (Tangent cone)**.** The *tangent cone* to Ω at $x \in \Omega$ is

$$T_\Omega(x) = \Big\{ s \in \mathbb{R}^n : \exists\, z^k \in \Omega,\ t^k \downarrow 0,\ z^k \to x,\ s = \lim_{k\to\infty} \frac{z^k - x}{t^k} \Big\}.$$

Definition 7.2 (Linearized feasible directions)**.** The set of *linearized feasible directions* at $x \in \Omega$ is

$$\mathcal{F}(x) = \{s : \nabla c_i(x)^\top s = 0,\ i \in E;\ \nabla c_i(x)^\top s \geq 0,\ i \in I \cap \mathcal{A}(x)\}.$$

Lemma 7.3. *$T_\Omega(x) \subseteq \mathcal{F}(x)$.*

Proof. [Proof to be filled in.] □

7.3 Constraint qualifications

Definition 7.4 (LICQ)**.** (CP) satisfies the *Linear Independence Constraint Qualification (LICQ)* at $x \in \Omega$ if the active constraint gradients $\{\nabla c_i(x) : i \in \mathcal{A}(x)\}$ are linearly independent.

Definition 7.5 (Slater CQ)**.** A convex problem (CP) (with $c_E(x) = Ax - b$, c_i concave for $i \in I$) satisfies the *Slater CQ* if there exists $\bar{x}$ with $A\bar{x} = b$ and $c_i(\bar{x}) > 0$ for all $i \in I$.

Theorem 7.6 (CQ implies $T_\Omega = \mathcal{F}$)**.** *Under LICQ (or any standard CQ), $T_\Omega(x) = \mathcal{F}(x)$.*

Proof. [Proof to be filled in.] □

7.4 First-order necessary conditions (KKT)

Definition 7.7 (Lagrangian)**.** The *Lagrangian* of (CP) is

$$\mathcal{L}(x, y, \lambda) = f(x) - \sum_{i \in E} y_i c_i(x) - \sum_{i \in I} \lambda_i c_i(x).$$

Theorem 7.8 (KKT first-order necessary conditions)**.** *Let x^* be a local minimizer of* (CP) *at which a CQ holds. Then there exist multipliers $y^* \in \mathbb{R}^{|E|}$ and $\lambda^* \in \mathbb{R}^{|I|}$ such that*

$$\begin{aligned}
\nabla f(x^*) &= \sum_{i \in E} y_i^* \nabla c_i(x^*) + \sum_{i \in I} \lambda_i^* \nabla c_i(x^*), \\
c_i(x^*) &= 0, \quad i \in E, \\
c_i(x^*) &\geq 0, \quad i \in I, \\
\lambda_i^* &\geq 0, \quad i \in I, \\
\lambda_i^* c_i(x^*) &= 0, \quad i \in I \quad \textit{(complementarity)}.
\end{aligned}$$

Proof. [Proof to be filled in.] □

7.5 Second-order conditions

Let $\mathcal{F}(\lambda^*) = \{s \in \mathcal{F}(x^*) : \nabla c_i(x^*)^\top s = 0 \text{ for } i \in I \text{ with } \lambda_i^* > 0\}$.

Theorem 7.9 (Second-order necessary conditions)**.** *Let x^* be a local minimizer of* (CP) *at which a CQ holds, with multipliers (y^*, λ^*). Then $s^\top \nabla_{xx}^2 \mathcal{L}(x^*, y^*, \lambda^*) s \geq 0$ for all $s \in \mathcal{F}(\lambda^*)$.*

Proof. [Proof to be filled in.] □

Theorem 7.10 (Second-order sufficient conditions)**.** *Let x^* be feasible for* (CP)*, and let (y^*, λ^*) satisfy the KKT conditions at x^*. If $s^\top \nabla_{xx}^2 \mathcal{L}(x^*, y^*, \lambda^*) s > 0$ for all $0 \neq s \in \mathcal{F}(\lambda^*)$, then x^* is a strict local minimizer of* (CP)*.*

Proof. [Proof to be filled in.] □

7.6 Convex problems

Theorem 7.11 (KKT is sufficient for convex problems)**.** *Suppose* (CP) *is a convex problem (i.e. f convex, c_i concave for $i \in I$, c_E affine). If $\hat{x}$ is a KKT point of* (CP)*, then $\hat{x}$ is a global minimizer of* (CP)*.*

Proof. [Proof to be filled in.] □

Chapter 8

Penalty Methods

We now turn to algorithms for the equality-constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad c(x) = 0, \tag{eCP}$$

where $c = (c_1, \ldots, c_m) : \mathbb{R}^n \to \mathbb{R}^m$ is smooth.

8.1 The quadratic penalty function

Definition 8.1 (Quadratic penalty function)**.** For $\sigma > 0$, the *quadratic penalty function* associated with (eCP) is

$$\Phi_\sigma(x) = f(x) + \frac{1}{2\sigma} \|c(x)\|^2 . \tag{8.1}$$

Algorithm 8.2 (Basic quadratic penalty method)**.** Choose $\sigma_0 > 0$ and $\varepsilon_0 > 0$. For $k = 0, 1, 2, \ldots$:

1. Find x^k with $\left\|\nabla \Phi_{\sigma_k}(x^k)\right\| \le \varepsilon_k$.
2. Choose $\sigma_{k+1} < \sigma_k$ and $\varepsilon_{k+1} < \varepsilon_k$.

Lemma 8.3 (Gradient of Φ_σ)**.** *With $J(x)$ the Jacobian of c,*

$$\nabla \Phi_\sigma(x) = \nabla f(x) + \frac{1}{\sigma} J(x)^\top c(x) = \nabla f(x) - J(x)^\top y(x),$$

where $y(x) = -c(x)/\sigma$.

Proof. [Proof to be filled in.] □

8.2 Convergence of the penalty method

Theorem 8.4 (Global convergence of the quadratic penalty method)**.** *Apply Algorithm 8.2 to* (eCP)*. Assume $f, c \in \mathcal{C}^1$, set $y_i^k = -c_i(x^k)/\sigma_k$, and assume $\left\|\nabla \Phi_{\sigma_k}(x^k)\right\| \le \varepsilon_k$ with $\varepsilon_k \to 0$ and $\sigma_k \to 0$ as $k \to \infty$. Suppose further that $x^k \to x^*$ where $\{\nabla c_i(x^*) : i = 1, \ldots, m\}$ are linearly independent. Then x^* is a KKT point of* (eCP) *and $y^k \to y^*$, the corresponding Lagrange multiplier vector.*

Proof. [Proof to be filled in.] □

Remark 8.5. If $J(x^*)$ is not full rank then x^* instead (locally) minimizes the infeasibility $\|c(x)\|$.

8.3 Other penalty functions

Definition 8.6 (Exact penalty function)**.** A penalty function $\Phi(x, \sigma)$ is *exact* if there exists $\sigma^* > 0$ such that for all $\sigma < \sigma^*$, every local solution of the original problem is a local minimizer of $\Phi(\cdot, \sigma)$.

Example 8.7 (ℓ_1 and ℓ_2 penalty functions)**.** For (CP),

$$\Phi_1(x, \sigma) = f(x) + \tfrac{1}{\sigma} \sum_{i \in E} |c_i(x)| + \tfrac{1}{\sigma} \sum_{i \in I} [c_i(x)]_-,$$

$$\Phi_2(x, \sigma) = f(x) + \tfrac{1}{\sigma} \left\| c_E(x) \right\|_2 + \tfrac{1}{\sigma} \left\| [c_I(x)]_- \right\|_2 .$$

Both are exact (but generally non-smooth); the quadratic penalty is smooth but not exact.

Chapter 9

Augmented Lagrangian Methods

9.1 The augmented Lagrangian

Definition 9.1 (Augmented Lagrangian)**.** The *augmented Lagrangian* for (eCP) is

$$\Phi(x, u, \sigma) = f(x) - u^\top c(x) + \frac{1}{2\sigma} \|c(x)\|^2 , \tag{9.1}$$

where $u \in \mathbb{R}^m$ is an estimate of the Lagrange multipliers and $\sigma > 0$.

Remark 9.2. $\Phi(x, u, \sigma)$ has two interpretations: (i) the Lagrangian augmented with a quadratic penalty; (ii) a shifted quadratic penalty with shift u.

9.2 The basic algorithm

Algorithm 9.3 (Augmented Lagrangian method)**.** Choose $u^0 \in \mathbb{R}^m$, $\sigma_0 > 0$, $\varepsilon_0 > 0$. For $k = 0, 1, \ldots$:

1. Find x^k with $\left\|\nabla_x \Phi(x^k, u^k, \sigma_k)\right\| \leq \varepsilon_k$.
2. Update $u^{k+1} = u^k - c(x^k)/\sigma_k$.
3. Choose $\sigma_{k+1} \leq \sigma_k$, $\varepsilon_{k+1} < \varepsilon_k$.

9.3 Convergence

Theorem 9.4 (Global convergence of augmented Lagrangian)**.** *Apply Algorithm 9.3 to* (eCP)*. Assume $f, c \in \mathcal{C}^1$, let $y^k = u^k - c(x^k)/\sigma_k$, and $\left\|\nabla_x \Phi(x^k, u^k, \sigma_k)\right\| \leq \varepsilon_k$ with $\varepsilon_k \to 0$. Suppose $x^k \to x^*$ where $\{\nabla c_i(x^*)\}_{i=1}^m$ are linearly independent. Then $y^k \to y^*$ with $\nabla f(x^*) - J(x^*)^\top y^* = 0$. Moreover, if either $\sigma_k \to 0$ for bounded u^k, or $u^k \to y^*$ for bounded σ_k, then x^* is a KKT point of* (eCP) *with associated multipliers y^*.*

Proof. [Proof to be filled in.] □

Chapter 10

Interior Point Methods

We consider the inequality-constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad c(x) \geq 0, \tag{iCP}$$

where $c = (c_1, \ldots, c_p) : \mathbb{R}^n \to \mathbb{R}^p$ is smooth.

10.1 The logarithmic barrier function

Definition 10.1 (Barrier function)**.** For $\mu > 0$, the *logarithmic barrier function* is

$$f_\mu(x) = f(x) - \mu \sum_{i=1}^{p} \log c_i(x), \tag{10.1}$$

defined on the strict interior $\{x : c(x) > 0\}$.

Lemma 10.2 (Barrier derivatives)**.** *With $J(x)$ the Jacobian of c and $C(x) = \operatorname{diag}(c_1(x), \ldots, c_p(x))$,*

$$\nabla f_\mu(x) = \nabla f(x) - \mu J(x)^\top c^{-1}(x) = \nabla f(x) - J(x)^\top \lambda(x),$$

$$\nabla^2 f_\mu(x) = \nabla^2 f(x) - \sum_{i=1}^{p} \frac{\mu}{c_i(x)} \nabla^2 c_i(x) + \mu J(x)^\top C(x)^{-2} J(x),$$

where $\lambda_i(x) := \mu / c_i(x)$ are estimates of the Lagrange multipliers.

Proof. [Proof to be filled in.] □

10.2 The basic barrier algorithm

Algorithm 10.3 (Basic barrier algorithm)**.** Choose $\mu_0 > 0$, $\varepsilon_0 > 0$ and x^0 with $c(x^0) > 0$. For $k = 0, 1, \ldots$:

1. Find x^k with $c(x^k) > 0$ and $\left\| \nabla f_{\mu_k}(x^k) \right\| \leq \varepsilon_k$.
2. Choose $\mu_{k+1} < \mu_k$, $\varepsilon_{k+1} < \varepsilon_k$.

10.3 Convergence

Theorem 10.4 (Global convergence of the barrier method)**.** *Apply Algorithm 10.3 to* (iCP)*. Assume $f, c \in \mathcal{C}^1$, $\lambda_i^k = \mu_k / c_i(x^k)$ for $i = 1, \ldots, p$ with $c(x^k) > 0$, $\mu_k > 0$, and $\left\| \nabla f_{\mu_k}(x^k) \right\| \leq \varepsilon_k$ with $\varepsilon_k \to 0$, $\mu_k \to 0$ as $k \to \infty$. Suppose $x^k \to x^*$ where the active gradients $\{\nabla c_i(x^*) : i \in \mathcal{A}\}$ are linearly independent (i.e. LICQ), $\mathcal{A} = \{i : c_i(x^*) = 0\}$. Then x^* is a KKT point of* (iCP) *and $\lambda^k \to \lambda^*$, the associated Lagrange multipliers.*

Proof. [Proof to be filled in.] □

10.4 Primal-dual interior point methods

Remark 10.5. Modern interior point software (KNITRO, IPOPT, GALAHAD) uses *primal-dual* variants that apply Newton's method directly to the perturbed KKT system

$$\nabla f(x) - J(x)^\top \lambda = 0, \qquad C(x)\lambda = \mu e, \qquad c(x) > 0,\ \lambda > 0,$$

where $e = (1, \ldots, 1)^\top$. The path traced out by solutions as $\mu \downarrow 0$ is called the *central path*.

Appendix A

Useful background

A.1 Taylor's theorem

Theorem A.1 (First-order Taylor)**.** *Let $f \in \mathcal{C}^1$. For any $x, s \in \mathbb{R}^n$ there exists $\tilde{\alpha} \in (0,1)$ with $f(x+s) = f(x) + \nabla f(x + \tilde{\alpha} s)^\top s$.*

Theorem A.2 (Second-order Taylor)**.** *Let $f \in \mathcal{C}^2$. For any $x, s \in \mathbb{R}^n$ there exists $\tilde{\alpha} \in (0,1)$ with*

$$f(x+s) = f(x) + \nabla f(x)^\top s + \tfrac{1}{2} s^\top \nabla^2 f(x + \tilde{\alpha} s) s.$$

A.2 Lipschitz continuity

Definition A.3. A function $g : \mathbb{R}^n \to \mathbb{R}^m$ is *Lipschitz continuous* on $\mathbb{R}^n$ with constant $L > 0$ if $\|g(y) - g(x)\| \leq L \|y - x\|$ for all $x, y \in \mathbb{R}^n$.

A.3 Convexity

Theorem A.4 (First-order characterization)**.** *Let $f \in \mathcal{C}^1$. Then f is convex if and only if $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$ for all x, y.*

Theorem A.5 (Second-order characterization)**.** *Let $f \in \mathcal{C}^2$. Then f is convex if and only if $\nabla^2 f(x) \succeq 0$ for all x.*